

Souham Ghosh

AI engineering lead | Model Optimization, On-device AI, Team Leadership

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Summary

AI engineering lead with 5+ years shipping production model optimization and on-device AI enablement for enterprise customers – quantization tooling (PTQ/QAT), custom Triton/CUDA kernels, and LLM post-training (SFT/DPO/GRPO) on constrained hardware. Currently leads a 10-person engineering team spanning development and delivery: owns technical roadmaps, code-review standards, mentoring and upskilling, and cross-team coordination with customer engineering organizations. Proven outcomes include 16× latency reductions and 200+ on-device deployments enabled; focused on scaling teams, delivery processes, and AI strategy.

Experience

Multicoreware Inc.

Jun 2022 – Present

Lead Associate – Model Optimization & On-device AI Enablement

Chennai

Project Lead – On-device Model Enablement Platform

Oct 2024 – Present

- Lead a 10-person engineering team delivering an on-device AI platform for the customer, enabling 200+ CV and VLA models through export, compile, profile, and quantize pipelines – 12 enablement reports shipped to production.
- Established code-review standards and Agile delivery cadence that kept every committed milestone on schedule; mentored engineers on quantization workflows and ONNX export constraints.
- Partner with the customer's engineering organization as the primary technical interface, translating model-enablement requirements into coordinated cross-team deliverables.
- Drive framework-level optimization for 200+ CV and VLA models – ONNX export graph constraints, static-vs-dynamic op handling, FP16/INT8 quantization schedules – with on-device latency and accuracy held within customer targets.

Senior Software Engineer – BEV/GKT Perception Flagship

May 2023 – Oct 2024

- Owned end-to-end performance for a production multi-camera BEV perception stack (BEVFormer-class) for a customer: rebuilt multi-scale deformable attention and spatial-temporal query grid sampling from scratch as custom PyTorch operators with full ONNX exportability, reducing end-to-end inference latency from 800 ms to 50 ms (16×) at preserved accuracy – via graph-level rewrites, operator fusion, and memory-layout tuning.
- Diagnosed and removed framework-level bottlenecks across a 3D perception pipeline: static-vs-dynamic ONNX export constraints, attention operator fusion, and memory-access-pattern tuning, validated against on-target benchmarks.
- Cut latency of a grid-sampling transformer (GKT) for BEV perception from 1.4 s to 59 ms at preserved accuracy by restructuring sampling and attention passes at the framework level.

Software Engineer – Model Quantization Tooling

Jun 2022 – Apr 2023

- Engineered and maintained a model quantization zoo (PTQ/QAT, W8A8/W4A8) covering 30+ models across PyTorch and TensorFlow, delivering reproducible quantization workflows for a customer.
- Built evaluation harnesses and CI workflows for accuracy and latency regression across quantization configurations, adopted by the wider team.
- Authored model-zoo documentation and trained new team members on the workflows, making the tooling the team's default path for 30+ model deliveries.

Rebrewed Technologies

Oct 2021 – Apr 2022

Data Scientist (Part-Time)

Remote

- Directed a team of 8 to deliver a customized portal for a pharmaceutical client that automated product web hosting using Computer Vision and NLP across 5,260+ product images.
- Automated the web-hosting pipeline end-to-end – U2-Net background removal, YOLOv5 in-image detection, super-resolution enhancement, and Pytesseract OCR auto-fill of illegible text fields – with 83% of fields auto-filled correctly.
- Implemented content-based filtering recommendation systems for hosted websites from scratch.

Cognizant Technology Solutions

2020 – May 2022

Programmer Analyst

Kolkata

- Shipped production enhancements to a power utility's customer-facing web platform (Spring Boot, Vue.js) as part of a 10-member team.

NCFlexe, IIT Kanpur

Intern

Aug 2020 – Nov 2020

Remote

- Trained a cattle identification model based on muzzle patterns using One-Shot Learning on a 5,000-image dataset.
- Generated synthetic cattle data with a CycleGAN, expanding the dataset to 10,000 images.
- Delivered the application as a mobile app.

Skills

Leadership & Delivery: leading cross-functional engineering teams (10 people), mentoring and upskilling, technical roadmapping, code-review standards, customer and stakeholder coordination, cross-team delivery planning

Model Optimization: PTQ & QAT, W8A8 / W4A8 quantization, FP16/INT8 schedules, knowledge distillation, ONNX export constraints

Custom Kernels & Compilers: OpenAI Triton, CUDA, TVM, MLIR, NCU/NSYS profiling, roofline analysis

LLM Post-Training & RL: SFT, DPO, GRPO, PPO, verifiable rewards, reasoning training

Programming Languages: Python, PyTorch, TensorFlow, C++, Git/CI

Education

Institute of Engineering & Management

2016 – 2020

Bachelor of Technology in Information Technology (GPA: 7.67 / 10.00)

Kolkata, West Bengal

Projects

Triton Fused Multi-Scale Deformable-Attention Kernel | OpenAI Triton, PyTorch

- Wrote a block-tiled Triton kernel for multi-scale deformable attention, fusing bilinear sampling and level accumulation into a single kernel launch, with numerical validation against the PyTorch baseline (torch.allclose).
- Benchmarked the attention evolution from naive PyTorch (2.80 ms) to autotuned Triton (0.74 ms), with NCU/NSYS profiling and roofline analysis.

GRPO Reasoning Trainer (Qwen2.5-1.5B) | PyTorch, Transformers, SymPy

- Developed a custom Group Relative Policy Optimization (GRPO) trainer on Qwen2.5-1.5B with SymPy-verifiable GSM8K rewards (group size G=4), including a DPO baseline for comparison.
- Tracked reward, response length, and accuracy curves during training, capturing the emergence of self-correction.

Publications

Unsupervised Learning Based Evaluation of Player Performances (2021) – Innovations in Systems and Software Engineering, vol. 17, pp. 121–130.

Framework for Appraisal of Twenty-Twenty League Player (2020) – Proceedings of the Global AI Congress 2019, pp. 671–679, Advances in Intelligent Systems and Computing (AISC), vol. 1112.